



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE
NASIONALE
SENIOR SERTIFIKAAT**

GRADE 12/GRAAD 12

TECHNICAL MATHEMATICS P2/TEGNIESE WISKUNDE V2

NOVEMBER 2024

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
M	Method/Metode
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
F	Formula/Formule
SF	Substitution in correct formula/Vervanging in korrekte formule
ST	Statement/Bewering
ST/RE	Statement with reason/Bewering met rede

**These marking guidelines consist of 27 pages.
Hierdie nasienriglyne bestaan uit 27 bladsye.**

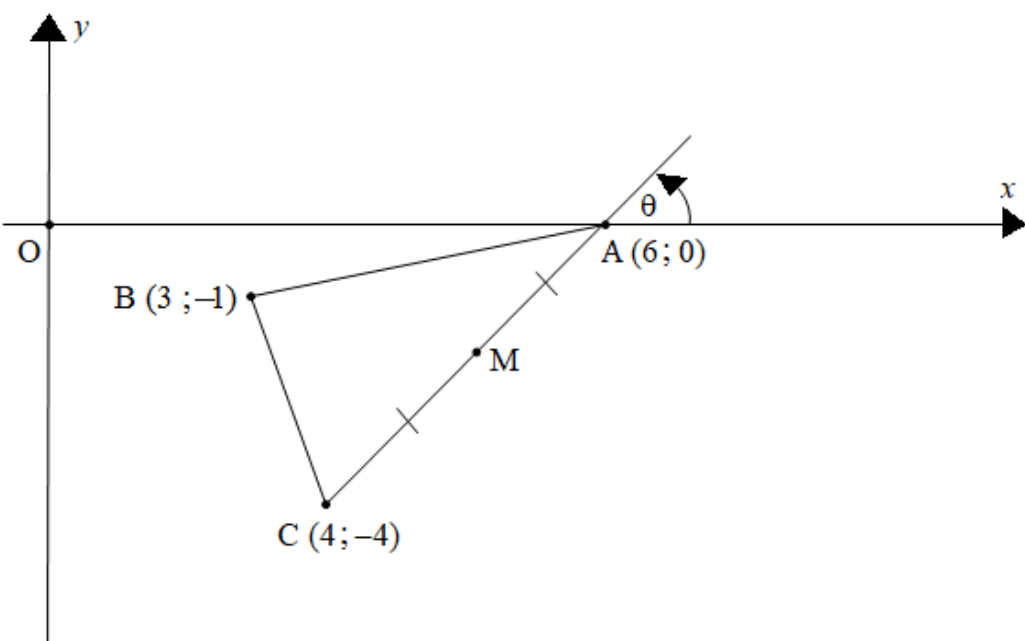
NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- The method of consistent accuracy marking must be applied in all aspects of the marking guidelines where applicable as indicated by the marking code CA.

LET WEL:

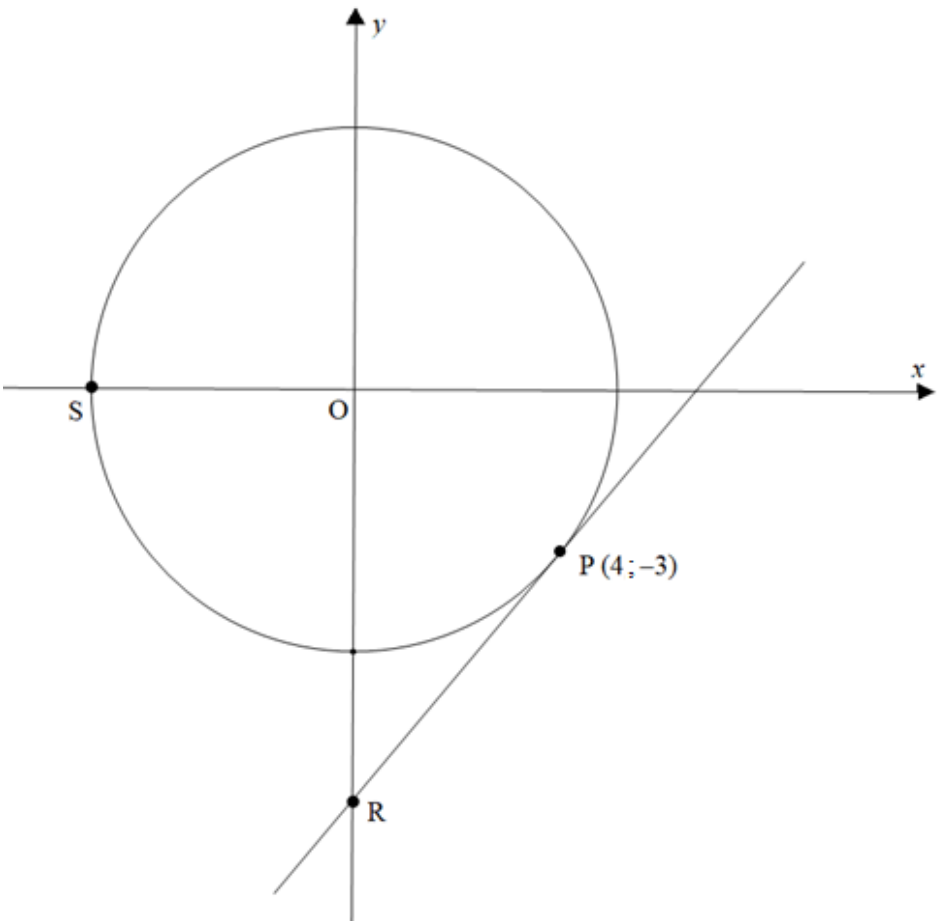
- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Die metode van volgehoue akkuraatheid-nasien moet waar moontlik op alle aspekte van die nasienriglyne toegepas word, soos aangedui deur die nasienkode CA.

QUESTION/VRAAG 1

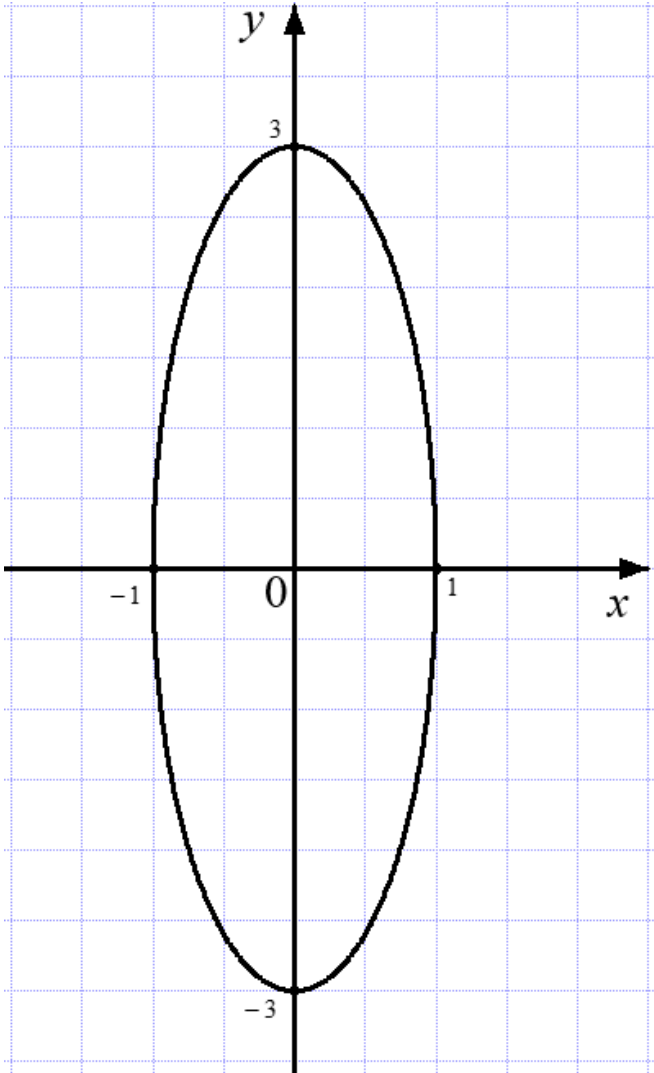
		
1.1	$m_{AC} = \frac{y_C - y_A}{x_C - x_A}$ $= \frac{-4 - 0}{4 - 6}$ $= 2$	✓ SF A ✓ gradient / gradiënt CA (2) AO: Full marks
1.2	$\tan \theta = m_{AC}$ $\theta = \tan^{-1}(2)$ $\theta \approx 63,43^\circ$	✓ SF CA ✓ value of / waarde van θ CA (2) AO: Full marks

1.3	$BC = \sqrt{(x_B - x_C)^2 + (y_B - y_C)^2}$ $= \sqrt{(3 - 4)^2 + (-1 - (-4))^2}$ $= \sqrt{10} \approx 3,16$	✓SF A ✓ length/ <i>lengte</i> BC CA (2) AO: Full marks
1.4	$M \left(\frac{x_A + x_C}{2}; \frac{y_A + y_C}{2} \right)$ $M \left(\frac{6 + 4}{2}; \frac{0 + (-4)}{2} \right)$ $M (5; -2)$	✓ x-coordinate/<i>koördinaat</i> A ✓ y-coordinate/<i>koördinaat</i> A (2)
1.5	$m_{\text{perp. bisector}} = -\frac{1}{2}$ $y - (-2) = -\frac{1}{2}(x - 5) \text{ OR/OF } -2 = -\frac{1}{2}(5) + c$ $y = -\frac{1}{2}x + \frac{5}{2} - 2 \qquad c = \frac{5}{2} - 2$ $\therefore y = -\frac{1}{2}x + \frac{1}{2}$	✓ <i>m</i> of perp bisect/ <i>van loodregte Middellyn</i> CA ✓ substitution/ <i>vervang</i> <i>m</i> CA ✓ substitution/ <i>vervang</i>(5 ; -2) CA ✓ equation in form <i>y = ...</i> / <i>vergelyking in vorm y = ...</i> CA (4)
		[12]

QUESTION/VRAAG 2

2.1			
2.1.1	$x^2 + y^2 = r^2$ $(4)^2 + (-3)^2 = r^2$ $r^2 = 25$ $\therefore x^2 + y^2 = 25$ OR/OF $\therefore x = \pm \sqrt{25 - y^2}$ OR/OF $y = \pm \sqrt{25 - x^2}$ OR/OF $x^2 + y^2 = (4)^2 + (-3)^2$ $= 25$	✓ SF A ✓ equation/vergelyking CA OR/OF ✓ SF A ✓ equation/vergelyking CA (2) AO: Full marks	
2.1.2	$S(-5; 0)$	✓ -5 CA ✓ 0 A (2)	

2.1.3	$m_{OP} = -\frac{3}{4}$ $m_{PR} = \frac{4}{3}$ $y - (-3) = \frac{4}{3}(x - 4) \quad \text{OR/OF} \quad -3 = \frac{4}{3}(4) + c$ $y = \frac{4}{3}x - \frac{16}{3} - 3 \quad c = -\frac{16}{3} - 3$ $\therefore y = \frac{4}{3}x - \frac{25}{3}$ OR/OF $x \cdot x_1 + y \cdot y_1 = r^2$ $4x - 3y = 25$ $-3y = -4x + 25$ $y = \frac{4}{3}x - \frac{25}{3}$	✓ gradient/gradjënt of/van OP A ✓ gradient/gradjënt of/van PR CA ✓ subst / vervang (4; -3) A ✓ equation/ vergelyking CA OR/OF ✓ F A ✓ subst / vervang (4; -3) A ✓ subst / vervang r^2 CA ✓ equation /vergelyking CA (4)
2.1.4	$y = -\frac{25}{3} \approx -8,33 \quad \text{OR/OF} \quad \left(0; -\frac{25}{3}\right)$	✓ y-intercept/afsnit CA (1)

2.2.1	$\frac{x^2}{1^2} + \frac{y^2}{3^2} = 1$	✓ standard form / <i>std vorm</i> A (1)
2.2.2		✓ x- and y-intercepts/ <i>afsnitte</i> CA ✓ elliptical shape/ <i>elliptiese vorm</i> CA (2)
		[12]

QUESTION/VRAAG 3

3.1.1	$A = \frac{17}{60} \pi = 51^\circ$	✓ A in degrees / <i>grade</i> A (1)
3.1.2	$\sqrt{\operatorname{cosec} B}$ $= \sqrt{\operatorname{cosec} 34^\circ}$ OR/OF $= \sqrt{\operatorname{cosec} \left(34^\circ \times \frac{\pi}{180} \right)}$ $= \sqrt{\frac{1}{\sin 34^\circ}}$ $= \sqrt{\operatorname{cosec} \frac{17}{90} \pi}$ $\approx 1,34$	✓ substitution / <i>vervang</i> A ✓ S CA (2)
3.1.3	$\tan(A + B)$ $= \tan(51^\circ + 34^\circ)$ OR / OF $= \tan \left(85^\circ \times \frac{\pi}{180} \right)$ $\approx 11,43$	✓ substitution / <i>vervang</i> CA ✓ S CA (2)

3.2.1	$\sec \theta = -\frac{13}{5}$ OR / OF $\frac{1}{\cos \theta} = \frac{1}{\frac{-5}{13}}$	✓ ratio / verhouding	A (1)
3.2.2	$x^2 + y^2 = r^2$ $(-5)^2 + n^2 = (13)^2$ $n^2 = 144$ $n = 12$ $1 + \sin^2 \theta$ $= 1 + \left(\frac{12}{13}\right)^2$ $= \frac{313}{169}$ OR/OF $1 + \sin^2 \theta = 1 + 1 - \cos^2 \theta$ $= 2 - \cos^2 \theta$ $= 2 - \left(\frac{-5}{13}\right)^2$ $= \frac{313}{169}$	✓ SF ✓ value of/waarde van n ✓ sin ratio / verh ✓ S OR/OF ✓ I ✓ $2 - \cos^2 \theta$ ✓ $\frac{-5}{13}$ ✓ S	A CA CA CA A CA CA CA CA (4)

3.2.2	$x^2 + y^2 = r^2$ $(-5)^2 + n^2 = (13)^2$ $n^2 = 144$ $n = 12$ $1 + \sin^2 \theta$ $= 1 + \left(\frac{12}{13}\right)^2$ $= \frac{313}{169}$ OR/OF $1 + \sin^2 \theta = 1 + 1 - \cos^2 \theta$ $= 2 - \cos^2 \theta$ $= 2 - \left(\frac{-5}{13}\right)^2$ $= \frac{313}{169}$	✓ SF A ✓ value of/waarde van n CA ✓ sin ratio / verh CA ✓ S CA OR/OF ✓ I A ✓ $2 - \cos^2 \theta$ CA ✓ $\frac{-5}{13}$ CA ✓ S CA (4)
3.3	$2 \sin x = 3 \cos x$ $\frac{2 \sin x}{2 \cos x} = \frac{3 \cos x}{2 \cos x}$ OR / OF $\frac{2 \sin x}{3 \sin x} = \frac{3 \cos x}{3 \sin x}$ $\tan x = \frac{3}{2}$ $\cot x = \frac{2}{3}$ Ref. angle /verw hoek = $56,31^\circ$ $x = 180^\circ + 56,31^\circ$ $\therefore x = 236,31^\circ$	✓ dividing by / deel deur $\cos x$ or $\sin x$ A ✓ tan/cot ratio / verh CA ✓ Ref. angle /verw hoek CA ✓ size of / grootte van x CA (4)
		[14]

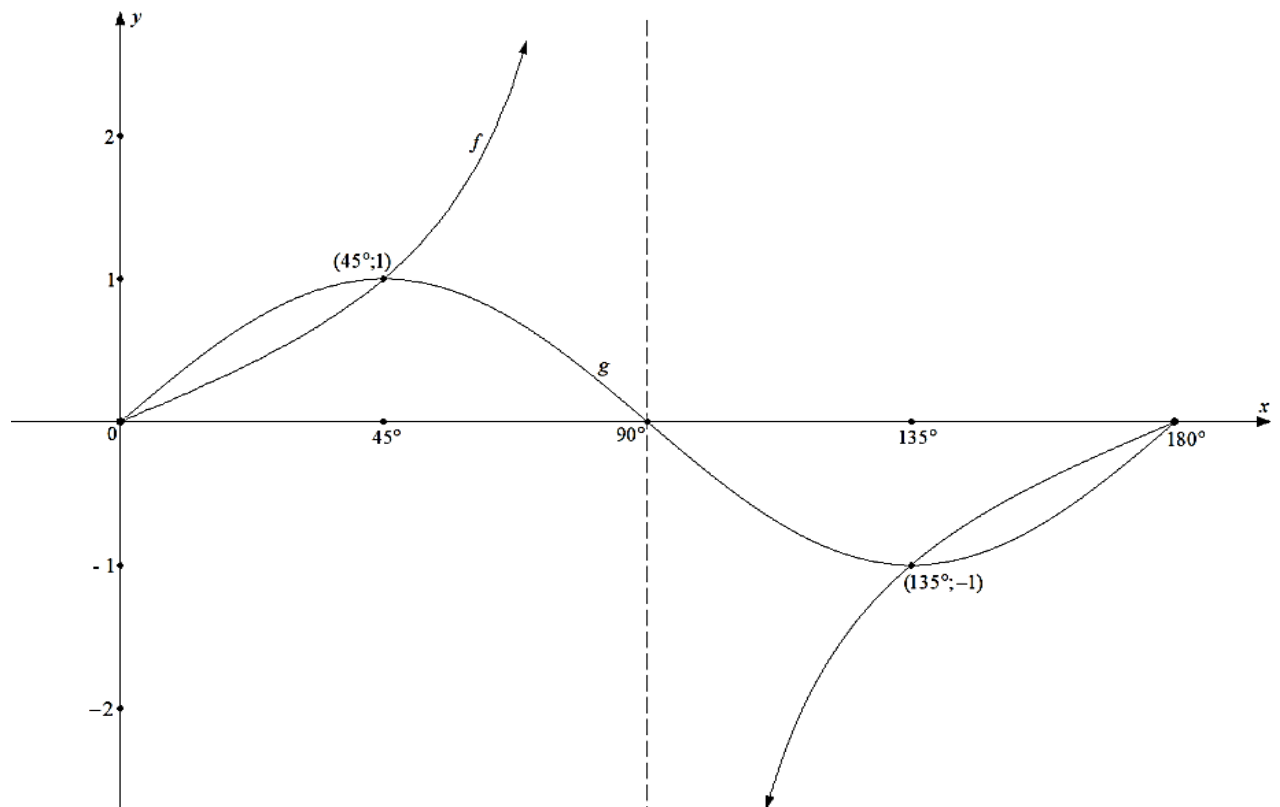
QUESTION/VRAAG 4

4.1.1	$-\sin x$	✓ reduction /reduksie A (1)
4.1.2	$-\cos x$	✓ reduction /reduksie A (1)
4.1.3	$\frac{\cot(180^\circ + x) \cdot \sin(2\pi - x)}{\cos(180^\circ - x) \cdot \cos(360^\circ - x) + 2\cos^2(180^\circ + x)}$ $= \frac{\cot x \cdot -\sin x}{-\cos x \cdot \cos x + 2(-\cos x)^2}$ $= \left(\frac{\cos x}{\sin x} \cdot -\frac{\sin x}{1} \right) \div (-\cos^2 x + 2\cos^2 x)$ $= \frac{-\cos x}{\cos^2 x}$ $= -\frac{1}{\cos x} \text{ OR / OF } -\sec x$	✓ $\cot x$ A ✓ $\cos x$ A ✓ $(-\cos x)^2 / \cos^2 x$ A ✓ cot quotient identity/ kwosiënt identiteit A ✓ S CA ✓ S CA (6)
4.2.1	$\cos^2 \theta$	✓ identity/identiteit A (1)
4.2.2	$\frac{1}{\sin \theta} - \frac{\sin \theta}{1 + \cos \theta} = \cot \theta$ $\text{L.H.S /LK.} = \frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta + \cos^2 \theta}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta(1 + \cos \theta)}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta}{\sin \theta} = \cot \theta = \text{RHS/RK}$ OR/OF $\text{L.H.S /LK.} = \frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta(1 + \cos \theta)}$ $= \frac{1 + \cos \theta - (1 - \cos^2 \theta)}{\sin \theta(1 + \cos \theta)}$ $= \frac{1 + \cos \theta - 1 + \cos^2 \theta}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta + \cos^2 \theta}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta(1 + \cos \theta)}{\sin \theta(1 + \cos \theta)}$ $= \frac{\cos \theta}{\sin \theta} = \cot \theta = \text{RHS/RK}$ OR/OF	✓ I $\cos^2 \theta$ A ✓ LCD/KGV A ✓ common factor / gemene faktor A ✓ S $\left(\frac{\cos \theta}{\sin \theta} \right)$ A OR/OF ✓ I $1 - \cos^2 \theta$ A ✓ LCD/KGV A ✓ common factor / gemene faktor A ✓ S $\left(\frac{\cos \theta}{\sin \theta} \right)$ A OR/OF

	$\begin{aligned} \text{LHS/LK} &= \frac{(1 + \cos \theta) - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \\ &= \frac{(1 + \cos \theta) - (1 - \cos^2 \theta)}{\sin \theta (1 + \cos \theta)} \\ &= \frac{(1 + \cos \theta) - [(1 + \cos \theta)(1 - \cos \theta)]}{\sin \theta (1 + \cos \theta)} \\ &= \frac{(1 + \cos \theta)[1 - (1 - \cos \theta)]}{\sin \theta (1 + \cos \theta)} \\ &= \frac{1 - 1 + \cos \theta}{\sin \theta} \\ &= \frac{\cos \theta}{\sin \theta} \\ &= \cot \theta = \text{RHS/RK} \end{aligned}$	✓ I $1 - \cos^2 \theta$ A ✓ factors/<i>faktore</i> A ✓ S A ✓ S $\left(\frac{\cos \theta}{\sin \theta} \right)$ A (4)
		[13]

QUESTION/VRAAG 5

5.1

For/vir f :

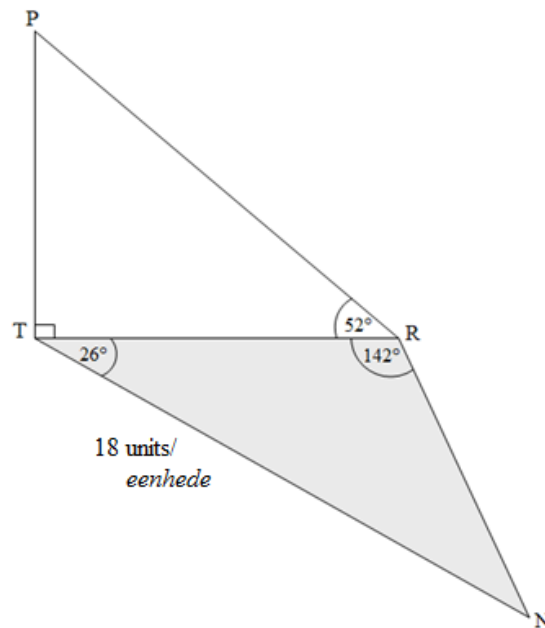
- ✓ shape / vorm **A**
- ✓ asymptote / asimptoot **A**
- ✓ intercepts / afsnitte **A**

For/vir g :

- ✓ shape / vorm **A**
- ✓ intercepts / afsnitte **A**
- ✓ turning points / draaipte **A**

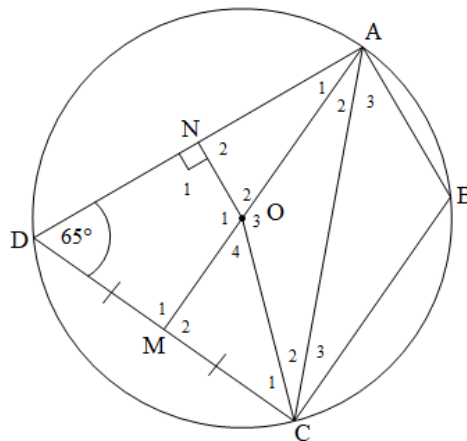
(6)

5.2.1	180°	✓ period / periode	CA (1)
5.2.2	$x = 0^\circ$ $x = 45^\circ$ $x = 135^\circ$ $x = 180^\circ$	Any Two/Enig twee ✓ x value/ waarde CA ✓ x value/ waarde CA	CA (2)
5.2.3	2	✓ amplitude	CA (1)
5.2.4	$g(x) = 5 \sin 4x$	✓ 5 A ✓ $\sin 4x$ A	A (2)
			[12]

QUESTION/VRAAG 6

6.1	$\hat{N} = 12^\circ$	✓ angle size/hoek grootte A (1)
6.2	$\frac{TR}{\sin \hat{N}} = \frac{TN}{\sin \hat{T} R N} \quad \text{OR / OF} \quad \frac{n}{\sin N} = \frac{r}{\sin \hat{T} R N}$ $\frac{TR}{\sin 12^\circ} = \frac{18}{\sin 142^\circ}$ $TR = \frac{18 \sin 12^\circ}{\sin 142^\circ}$ $\approx 6,08 \text{ units/eenhede}$	✓ sine rule /sinus reël A ✓ substitution in sine rule / vervanging in sine reël CA ✓ length of / lengte van TR CA (3)
6.3	$\tan 52^\circ = \frac{PT}{6,08}$ $PT = 6,08 \tan 52^\circ$ $\approx 7,78 \text{ units/eenhede}$ OR/OF $\hat{P} = 38^\circ$ $\tan 38^\circ = \frac{6,08}{PT}$ $PT = \frac{6,08}{\tan 38^\circ}$ $\approx 7,78 \text{ units}$ OR/OF	✓ tan ratio /verh CA ✓ length of / lengte van PT CA OR/OF ✓ tan ratio /verh CA ✓ length of / lengte van PT CA OR/OF

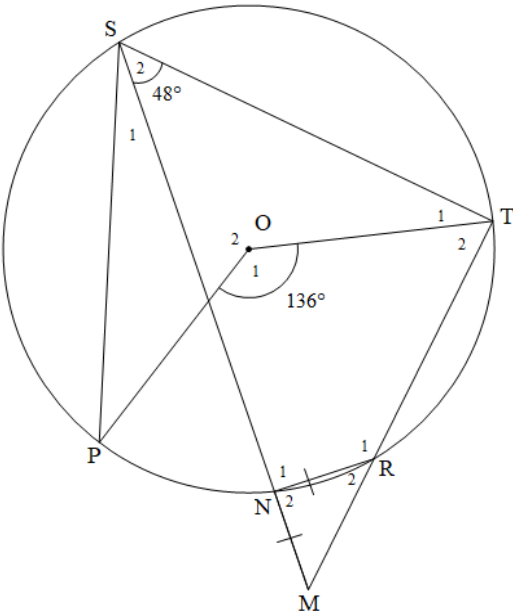
	$\frac{PR}{\sin 90^\circ} = \frac{6,08}{\sin 38^\circ}$ $PR = \frac{6,08 \cdot \sin 90^\circ}{\sin 38^\circ}$ $\approx 9,88 \text{ units}$ $PT = \sqrt{(9,88)^2 - (6,08)^2}$ $\approx 7,79 \text{ units}$	✓ length of PR CA ✓ length of / lengte van PT CA (2)
6.4	Area of/van Δ TRN = $\frac{1}{2} \times 18 \times 6,08 \times \sin 26^\circ$ $\approx 23,99$ Area of/van Δ PTR = $\frac{1}{2} \times 7,78 \times 6,08$ $\approx 23,65$ $\frac{\text{Area of/van } \Delta \text{ TRN}}{\text{Area of/van } \Delta \text{ PTR}} \approx \frac{23,99}{23,65} \approx 1,01$ OR/OF Area of/van Δ TRN = $\frac{1}{2} \times 18 \times 6,08 \times \sin 26^\circ$ $\approx 23,99$ Area of/van Δ PTR = $\frac{1}{2} \times 6,08 \times 9,87$ $\approx 23,64$ $\frac{\text{Area of/van } \Delta \text{ TRN}}{\text{Area of/van } \Delta \text{ PTR}} \approx \frac{23,99}{23,64} \approx 1,01$ OR/OF Area of/van Δ TRN = $\frac{1}{2} \times 18 \times 6,08 \times \sin 26^\circ$ $\approx 23,99$ Area of/van Δ PTR = $\frac{1}{2} \times 6,08 \times 7,78 \sin 90^\circ$ $\approx 23,65$ $\frac{\text{Area of/van } \Delta \text{ TRN}}{\text{Area of/van } \Delta \text{ PTR}} \approx \frac{23,99}{23,65} \approx 1,01$	✓ substitution /vervanging CA ✓ Area of/van Δ TRN CA ✓ substitution /vervanging CA ✓ Area of/van Δ PTR CA ✓ ratio / verhouding CA OR/OF ✓ substitution /vervanging CA ✓ Area of/van Δ TRN CA ✓ substitution /vervanging CA ✓ Area of/van Δ PTR CA ✓ ratio / verhouding CA OR/OF ✓ substitution /vervanging CA ✓ Area of/van Δ TRN CA ✓ substitution /vervanging CA ✓ Area of/van Δ PTR CA ✓ ratio / verhouding CA (5)
		[11]

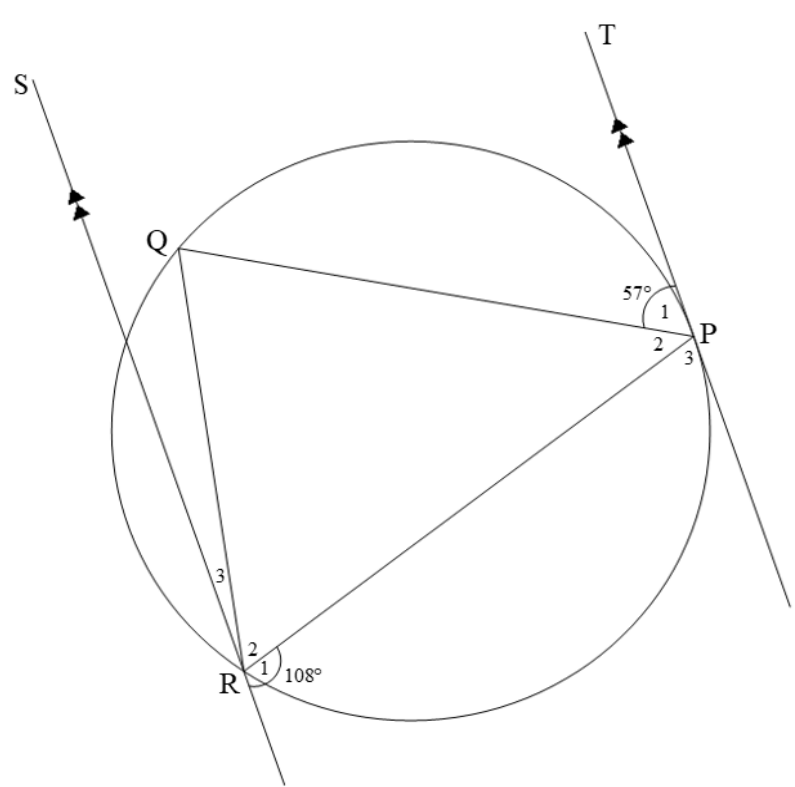
QUESTION/VRAAG 7


7.1	(line from centre to midpoint of chord / lyn vanaf midpt na mdpt van koord)		✓ RE	A (1)
7.2	(opp $\angle$'s supplementary / teenoorst $\angle$ 'e is supplementêr) OR/OF ext $\angle$ = opp interior $\angle$ / buite $\angle$ = teenoort binne $\angle$		✓ RE	A
7.3	Statement/ Bewering $\hat{O}_1 = 115^\circ$ $\hat{B} = 115^\circ$ DN = NA OR/OF $DN = \frac{1}{2} DA$	Reason / Rede Opp $\angle$ s of cyclic quad / Teenoorst $\angle$ e van kdvh Opp $\angle$ s of cyclic quad / Teenoorst $\angle$ e van kdvh Line from centre $\perp$ to chord Lyn vanaf midpt $\perp$ op koord	✓ ST ✓ ST ✓ ST ✓ RE	A A A A (4)

7.4	In $\triangle ADM$ & $\triangle ACM$: AM is common/ <i>gemeenskaplik</i> $\hat{M}_1 = 90^\circ = \hat{M}_2$ $\left(\begin{array}{l} \angle \text{ on str line / } \\ \angle \text{ op reguitlyn} \end{array} \right)$ DM = MC (given / <i>gegee</i>) $\therefore \triangle ADM \equiv \triangle ACM$ (S$\angle$S) OR/OF In $\triangle ADM$ & $\triangle ACM$: $\hat{M}_1 = \hat{M}_2 = 90^\circ$ $\left(\begin{array}{l} \text{line from cent. } \perp \text{ to chord / } \\ \text{lyn vanuit mdtp } \perp \text{ op koord} \end{array} \right)$ AM is common/ <i>gemeenskaplik</i> AD = AC (Pyth. theorem/ <i>stelling</i>) $\therefore \triangle ADM \equiv \triangle ACM$ (RHS/ <i>RSS</i>)	✓ ST A ✓ ST A ✓ RE (conclusion) A OR/OF ✓ ST A ✓ ST A ✓ RE A (3)
7.5	$\hat{O}_3 = 130^\circ$ $\left(\begin{array}{l} \angle \text{ at centre } = 2 \times \angle \text{ at circum / } \\ \text{midpts } \angle = 2 \times \text{ omtreks } \angle \end{array} \right)$ $\hat{O}_3 + \hat{B} = 130^\circ + 115^\circ \neq 180^\circ$ $\therefore ABCO$ is not cyclic quad. (opp $\angle$s not suppl) / $\therefore ABCO$ is nie 'n kvhk nie (teenoorst $\angle$e is nie sup pl) OR/OF Since three points A, B and C of the quadrilateral lie on the existing circle✓ and point O lies within the circle✓, therefore AOCB cannot be a cyclic quadrilateral✓. <i>Aangesien drie punte A, B en C van die vierhoek op die bestaande sirkel lê✓ en punt O binne die sirkel lê✓, kan AOCB dus nie 'n koordevierhoek wees nie✓.</i>	✓ ST A ✓ RE A ✓ RE A OR/OF ✓ ST A ✓ RE A ✓ RE (3)
		[12]

QUESTION/VRAAG 8

8.1			
8.1.1	$\hat{PST} = 68^\circ$ $\left(\begin{array}{l} \angle \text{ at centre} = 2 \times \angle \text{ at circum} / \\ \text{midpts } \angle = 2 \times \text{omtreks } \angle \end{array} \right)$ $\hat{S}_1 = 20^\circ$	$\checkmark$ ST $\checkmark$ RE $\checkmark$ ST	A A CA (3)
8.1.2	$\hat{R}_2 = 48^\circ$ $\left(\begin{array}{l} \text{ext } \angle \text{ of cyclic quad} / \\ \text{buite } \angle \text{ van kdvk} \end{array} \right)$ $\therefore \hat{M} = 48^\circ$ $\left(\begin{array}{l} \angle \text{s opp} = \text{sides} / \\ \angle \text{e teenoor} = \text{sye} \end{array} \right)$ $\therefore \hat{N}_2 = 84^\circ$ $\left(\begin{array}{l} \text{Int } \angle \text{s of } \Delta / \\ \text{Binne } \angle \text{e van } \Delta \end{array} \right)$ OR/OF $\hat{R}_1 = 132^\circ$ $\left(\begin{array}{l} \text{opp. } \angle \text{s of cyclic quad suppl.} / \\ \text{teenst } \angle \text{e van kvhk suppl} \end{array} \right)$ $\hat{R}_2 = 48^\circ$ $\left(\angle \text{s on a str. line} / \angle \text{op'n reguitlyn} \right)$ $\hat{M} = 48^\circ$ $\left(\angle \text{s opp.} = \text{sides} / \angle \text{e teenoor} = \text{sye} \right)$ $\hat{N}_2 = 84^\circ$ $\left(\angle \text{s of a } \Delta / \angle \text{e van } \Delta \right)$	$\checkmark$ ST $\checkmark$ RE $\checkmark$ ST $\checkmark$ RE $\checkmark$ ST $\checkmark$ RE $\checkmark$ ST $\checkmark$ RE $\checkmark$ ST	CA A CA A CA A CA A CA A A (6)

8.2				
8.2.1	$\hat{R}_2 = 57^\circ$	$\left(\begin{array}{l} \text{tan - chord thm /} \\ \text{raaklyn - koord st} \end{array} \right)$	✓ ST ✓ RE	A A (2)

8.2.2	$\therefore \hat{SRP} = 72^\circ$	$\left(\begin{array}{l} \angle \text{s on a str. line /} \\ \angle \text{eopreguitlyn} \end{array} \right)$	✓ ST	CA
	$\hat{P}_3 = 72^\circ$	$\left(\begin{array}{l} \text{co-int. } \angle \text{s; RS} \parallel \text{PT} \\ \text{ko - binne } \angle \text{s; RS} \parallel \text{PT} \end{array} \right)$	✓ ST ✓ RE	CA A
	$\hat{Q} = 72^\circ$	$\left(\begin{array}{l} \text{tan - chord th /} \\ \text{raaklyn - koord st} \end{array} \right)$	✓ ST	CA

	OR/OF $\hat{P}_2 + 57^\circ = 108^\circ = 51^\circ \quad \left(\begin{array}{l} \text{alt } \angle\text{s}; RS \parallel PT / \\ \text{verw. } \angle\text{e}; RS \parallel PT \end{array} \right)$ $\therefore \hat{Q} = 72^\circ \quad \left(\begin{array}{l} \text{Int } \angle\text{s of } \Delta / \\ \text{Binne } \angle\text{e van } \Delta \end{array} \right)$ $\hat{R}_3 = 15^\circ \quad \left(\begin{array}{l} \angle\text{s on str line} / \\ \angle\text{e op 'n reguit lyn} \end{array} \right)$ $\hat{SRP} = 15^\circ + 57^\circ = 72^\circ$ $\therefore \hat{SRP} = \hat{Q}$ OR/OF $\hat{P}_3 = 72^\circ \quad (\text{co-int./ ko-binne } \angle\text{s}; SR \parallel TP)$ $\hat{P}_2 = 51^\circ \quad (\angle\text{s on a str.line/op'n reguitlyn})$ $\hat{Q} = 72^\circ \quad (\text{int. } \angle\text{s of } \Delta / \text{binne } \angle\text{e van } \Delta)$ $\hat{SRP} = 72^\circ \quad (\text{int. } \angle\text{s of } \Delta / \text{binne } \angle\text{e van } \Delta)$ $\therefore \hat{SRP} = \hat{Q}$	OR/OF ✓ ST/RE A ✓ ST CA ✓ ST CA ✓ ST CA OR/OF ✓ ST/RE A ✓ ST CA ✓ ST CA ✓ ST CA (4)
		[15]

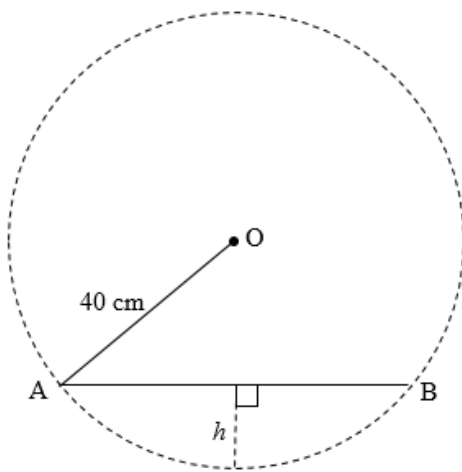
QUESTION/VRAAG 9

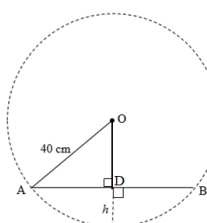
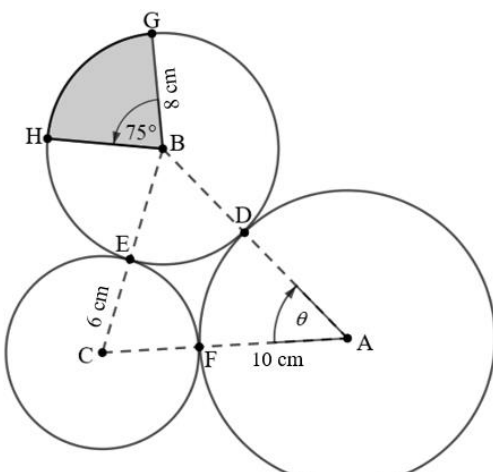
9.1	$VW = 6 \text{ cm}$	✓ ST	A (1)
9.2	$\frac{RP}{PS} = \frac{RV}{VW} \quad \left(\begin{array}{l} \text{prop th, } PV \parallel SW / \\ \text{ewerdig st; } PV \parallel SW \end{array} \right)$ OR/OF $\left(\begin{array}{l} \text{line } \parallel \text{ one side of } \Delta / \\ \text{lyn } \parallel \text{ een sy van } \Delta \end{array} \right)$ $\therefore \frac{RP}{9} = \frac{2}{6}$ $\therefore RP = 3 \text{ cm}$	✓ ST ✓ RE ✓ ST	A A CA (3)
9.3	$\frac{PV}{QW} = \frac{VT}{VW}$ $\frac{PV}{1,5} = \frac{12}{6}$ $PV = 3 \text{ cm}$ OR/OF $PQ = QT \quad \left(\begin{array}{l} \text{line from midpt. } \parallel \text{ to side} / \\ \text{lyn vanuit mdpt } \parallel \text{ aan sy} \end{array} \right)$ $PV = 2QW \quad \left(\begin{array}{l} \text{mid-point theorem} / \\ \text{mdpts stelling} \end{array} \right)$ $= 3 \text{ cm}$	✓ ST ✓ ST OR/OF ✓ ST ✓ ST AO: Full marks/volpunte	A A A A (2)

9.4	In Δs RPV and RSW: $\hat{R}$ is common / <i>gemeen</i> $\hat{P}_1 = \hat{S}_2$ $\left(\begin{array}{l} \text{corresp. } \angle\text{s; PV} \parallel \text{SW} / \\ \text{ooreenkst } \angle\text{e; PV} \parallel \text{SW} \end{array} \right)$ $\hat{V}_2 = \hat{W}_1$ $\left(\begin{array}{l} \text{corresp. } \angle\text{s; PV} \parallel \text{SW} / \\ \text{ooreenkst } \angle\text{e; PV} \parallel \text{SW} \end{array} \right)$ $\therefore \Delta\text{RPV} \parallel \Delta\text{RSW} \quad (\angle\angle\angle)$ OR/OF $\hat{R} = \hat{V}_2$ $(\angle\text{s opp.} = \text{sides} / \angle^e \text{ teenoor} = \text{sy})$ $\hat{V}_2 = \hat{W}_1$ $(\text{corr./ooreenk} \angle\text{s; PV} \parallel \text{SW})$ $\text{RS} = \text{SW}$ $(\text{sides opp.} = \angle\text{s} / \text{sy} \text{ teenoor} = \angle^e)$ $\frac{\text{RV}}{\text{RW}} = \frac{2}{8} = \frac{1}{4} \quad \frac{\text{PR}}{\text{RS}} = \frac{3}{12} = \frac{1}{4} \quad \frac{\text{PV}}{\text{SW}} = \frac{3}{12} = \frac{1}{4}$ $\therefore \Delta\text{RPV} \parallel \Delta\text{RSW} \quad (\text{corr.sides are in prop./ ooreenst sye in verh})$	✓ ST A ✓ ST A ✓ RE/3rd statement/3de stelling A OR/OF ✓ ST A ✓ ST A ✓ RE A (3)
9.5	$\therefore \frac{\text{SW}}{\text{RS}} = \frac{\text{PV}}{\text{RP}} \quad (\parallel \Delta\text{s})$ $\therefore \frac{\text{SW}}{12} = \frac{3}{3}$ $\therefore \text{SW} = 12$ OR / OF $\therefore \frac{\text{SW}}{\text{RW}} = \frac{\text{PV}}{\text{RV}} \quad (\parallel \Delta\text{s})$ $\therefore \frac{\text{SW}}{8} = \frac{3}{2}$ $\therefore \text{SW} = 12$ OR/OF	✓ ST CA ✓ ST CA OR / OF ✓ ST CA ✓ ST CA OR/OF

$\frac{RP}{RS} = \frac{PV}{SW}$ $\frac{3}{12} = \frac{3}{SW}$ $SW = 12$		✓ ST	CA
		✓ ST	CA
	OR/OF		
$\frac{PV}{SW} = \frac{RV}{RW}$ $\frac{3}{SW} = \frac{2}{8}$ $SW = 12$			OF/OF
		✓ ST	CA
		✓ ST	CA
	OR/OF		
$\hat{V}_2 = \hat{W}_1$ $\hat{V}_2 = \hat{R}$ $\hat{R} = \hat{W}_1$ $SR = SW$ $\therefore SR = 12$	(corr./ooreenk∠; PV SW) (∠s opp. = sides/) (∠e teenoor = sye)		OR/OF
		✓ ST	CA
		✓ ST	CA
			(2)
		AO: Full marks/volpunte	
			[11]

QUESTION/VRAAG 10

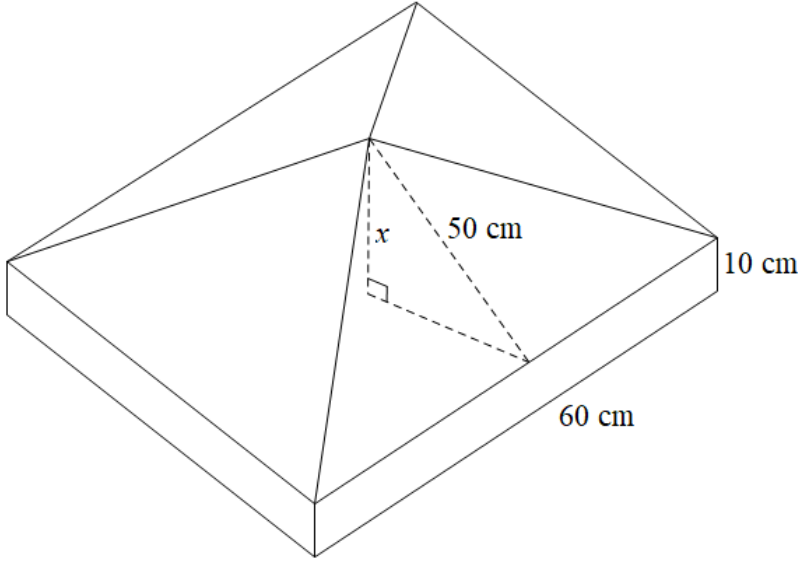
10.1			
10.1.1	$\frac{48}{60} = \frac{4}{5} = 0,8 \text{ rev/s}$	✓ $\frac{4}{5}$ or/of 0,8	A (1)
10.1.2	0,4 m	✓ 0,4	A (1)
10.1.3	0,8 m	✓ 0,8	CA (1)
10.1.4	$v = \pi D n$ $= \pi \times (0,8) \times (0,8)$ $= \frac{16}{25} \pi \text{ OR/OF } \approx 2,01 \text{ m/s}$ OR/OF $\omega = 2 \pi n$ $= 2 \pi \times (0,8) = \frac{8}{5} \pi \approx 5,03$ $v = \omega r$ $= \frac{8}{5} \pi \times 0,4$ $= \frac{16}{25} \pi \text{ OR/OF } \approx 2,01 \text{ m/s}$	✓ F ✓ SF ✓ circm vel /omtreksnld OR/OF ✓ F ✓ SF ✓ circm vel /omtreksnld	A CA CA A CA CA (3)
10.1.5	$4h^2 - 4dh + x^2 = 0$ $4(8)^2 - 4(80)(8) + x^2 = 0$ $x^2 = 2304$ $x = 48 \text{ cm}$ OR/OF	✓ F ✓ SF ✓ S ✓ value of / waarde van x OR/OF	A CA CA CA

	$OD = 40 - 8 = 32$ $AD = \sqrt{40^2 - 32^2}$ $AD = 24$ $DB = 24$ $AB = 48 \text{ cm}$		✓ length of/ <i>lengte van</i> OD ✓ substitution/ <i>vervanging</i> ✓ length of/ <i>lengte van</i> AD ✓ length of/ <i>lengte van</i> AB	A CA CA CA (4)
10.2				
10.2.1	$75^\circ = 75^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{12}$	OR/OF $\approx 1,31 \text{ rad}$	✓ angle/ <i>hoek in rad</i>	A (1)
10.2.2	Area of sector/ <i>Area van sektor</i> $= \frac{r^2 \theta}{2}$ $= \frac{(8)^2 \times \left(\frac{5\pi}{12}\right)}{2}$ OR/OF $\frac{(8)^2 \times (1,31)}{2}$ $= \frac{40\pi}{3} \approx 41,89 \text{ cm}^2$ OR/OF $\approx 41,92 \text{ cm}^2$ OR/OF Area of sector/ <i>Area van sektor</i> $= \frac{rs}{2}$ $= \frac{(8) \times \left(8 \times \frac{5\pi}{12}\right)}{2}$ OR/OF $\frac{(8) \times (8 \times 1,31)}{2}$ $= \frac{40\pi}{3} \approx 41,89 \text{ cm}^2$ OR/OF $\approx 41,92 \text{ cm}^2$ OR/OF	✓ F ✓ SF ✓ area OR/OF ✓ F ✓ SF ✓ area OR/OF	A CA CA A A CA	

	Area of sector/ $= \frac{\theta}{360^\circ} \pi r^2$ Area van sektor $= \frac{75^\circ}{360^\circ} \times \pi \times (8)^2$ $= \frac{40\pi}{3} \text{ cm}^2$ OR/OF $\approx 41,89 \text{ cm}^2$	✓ F A ✓ SF A ✓ area CA (3)
10.2.3	16 cm	✓ length / lengte A (1)
10.2.4	In ΔABC $\cos \theta = \frac{16^2 + 18^2 - 14^2}{2(16)(18)}$ $\theta = \cos^{-1} \left(\frac{16^2 + 18^2 - 14^2}{2(16)(18)} \right)$ $\theta = 48,19^\circ$ Arc length / Booglengte DF: $s = r \times \theta$ $s = (10) \times \left(48,19^\circ \times \frac{\pi}{180^\circ} \right)$ $= \frac{4819}{1800} \pi \text{ cm}$ OR/OF $\approx 8,41 \text{ cm}$ OR/OF In ΔABC $\cos \theta = \frac{16^2 + 18^2 - 14^2}{2(16)(18)}$ $\theta = \cos^{-1} \left(\frac{16^2 + 18^2 - 14^2}{2(16)(18)} \right)$ $\theta = 48,19^\circ$ $\frac{48,19^\circ}{360^\circ} = \frac{s}{20\pi}$ $\therefore s = \frac{48,19^\circ \times 20\pi}{360^\circ}$ $\approx 8,41 \text{ cm}$	✓ use Cosine rule / gebruik van kosinusreël A ✓ SF CA ✓ value of / waarde van θ CA ✓ F A ✓ arc length / booglengte CA OR/OF ✓ use Cosine rule / gebruik van kosinusreël A ✓ SF CA ✓ value of / waarde van θ CA ✓ M proportion / eweredigheid A ✓ arc length / booglengte CA (5)
		[20]

QUESTION/VRAAG 11

11.1			
11.1.1	$x = 0,8 \text{ m}$	✓ 0,8	A (1)
11.1.2	$a = 0,2 \text{ m}$	✓ $a = 0,2 \text{ m}$	CA (1)
11.1.3	$A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $= (0,2) \left(\frac{1,2 + 0,9}{2} + 1,15 + 1,25 + 1,1 \right)$ $= 0,91 \text{ m}^2$ OR/OF $A_T = a (m_1 + m_2 + m_3 + \dots + m_n)$ $= (0,2) \left(\frac{1,2 + 1,15}{2} + \frac{1,15 + 1,25}{2} + \frac{1,25 + 1,1}{2} + \frac{1,1 + 0,9}{2} \right)$ $= 0,91 \text{ m}^2$	✓ F ✓ SF ✓ area OR/OF ✓ F ✓ SF ✓ area	A CA CA A CA CA (3)
11.1.4	Area of Wall / <i>van muur</i> $= 3 \times 2 = 6 \text{ m}^2$ Area of un-plastered section / <i>Area van ongepleisterde gedeelte</i> $= 6 - 0,91 = 5,09 \text{ m}^2$ Cost to plaster rest of the wall / <i>Koste om die res van die muur te pleister</i> $= 5,09 \times 300$ $= \text{R } 1\,527$ $\text{R } 1700 > \text{R } 1527$ $\therefore$ the money will be sufficient	✓ Area of wall / <i>van muur</i> ✓ Area of un-plastered section / <i>Area van ongepleisterde gedeelte</i> ✓ M ✓ Cost / <i>Koste</i> ✓ Conclusion	A CA CA CA CA (5)

11.2			
11.2.1	$TSA = lb + 2lh + 2bh$ $= (60)(60) + 2(60)(10) + 2(60)(10)$ $= 6000 \text{ cm}^2$	✓ F ✓ SF ✓ Surface area / Buite-oppervl	A CA CA (3)
11.2.2	$x = \sqrt{50^2 - 30^2}$ $= 40 \text{ cm}$	✓ Pyth. ✓ length / lengte	A CA (2)
11.2.3	Vol. of rect. Prism / <i>regh prisma</i> $= 60 \times 60 \times 10 = 36\,000 \text{ cm}^3$ Vol. of pyramid / <i>van piramied</i> $= \frac{1}{3} \times (60 \times 60) \times 40 = 48\,000 \text{ cm}^3$ Total Vol. / <i>totale Vol</i> $= 36\,000 + 48\,000$ $= 84\,000 \text{ cm}^3$	✓ Vol. of rect. Prism / <i>van reghoekige prisma</i> ✓ Vol. of pyramid / <i>van piramied</i> ✓ Total Vol / <i>totale Vol</i>	A CA CA (3)
			[18]

TOTAL/TOTAAL: 150